Prevalence of anabolic steroid users seeking support from physicians: a systematic review and meta-analysis

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SUPPLEMENTARY MATERIAL

eTable 1: Search algorithms

PUBMED (results: 10,545)
((agents, anabolic[MeSH Terms]) OR (doping in sport[MeSH Terms]) OR (drugs, performance enhancing[MeSH Terms])) OR ((anabolic steroid*[Title/Abstract]) OR (anabolic androgenic[Title/Abstract]) OR (anabolic agent*[Title/Abstract]) OR (anabolic drug*[Title/Abstract]) OR (androgen* abuse[Title/Abstract]) OR (performance enhanc*[Title/Abstract]) OR (image enhanc*[Title/Abstract]) OR (synthetic androgen*[Title/Abstract]) OR (synthetic testosterone[Title/Abstract]) OR (testosterone ester*[Title/Abstract]) OR (designer steroid*[Title/Abstract]) OR (doping[Title/Abstract]) OR ("non prescript*" steroid*[Title/Abstract]) OR ("non medic*" steroid*[Title/Abstract]) OR ("non medic*" androgen*[Title/Abstract]) OR ("non prescript*" androgen*[Title/Abstract])) NOT ((review[Title/Abstract]) OR (animal*[Title/Abstract]) OR (bovine[Title/Abstract]) OR (in vitro[Title/Abstract]) OR (mice*[Title/Abstract]) OR (pig[Title/Abstract]) OR (pigs[Title/Abstract]) OR (rat[Title/Abstract]) OR (rats[Title/Abstract]) OR (mouse[Title/Abstract]) OR (spectrometry[Title/Abstract]) OR (nano*[Title/Abstract]) OR (conductor*[Title/Abstract]) OR (semiconductor[Title/Abstract]) OR (electro*[Title/Abstract]) OR (optic*[Title/Abstract]) OR (hydrogen[Title/Abstract]) OR (superconductor*[Title/Abstract]) OR (catalyst*[Title/Abstract]) OR (thermoelectric[Title/Abstract]) OR (ions[Title/Abstract])) AND (humans[Filter])
Web of Science (results: 1,261)
((TS=("doping in sport" OR "performance enhancing drug*"OR "anabolic steroid*" OR "anabolic androgenic*" OR "anabolic agent*" OR "anabolic drug*")) NOT (TS=(review OR animal* OR bovine OR “in vitro” OR mice* OR pig OR pigs OR rat OR rats OR mouse OR spectrometry OR nano* OR conductor* OR semiconductor OR electro* OR optic* OR hydrogen OR superconductor* OR catalyst* OR thermoelectric OR ions))) Filters: Articles, Sport Sciences, Medicine (all), Psychology (all), Toxicology
APA PsycNet (results: 816) A
((anabolic or doping or performance enhancing) not (review or animal or bovine or vitro or mice* or pig* or rat* or mouse or spectrometry or nano* or conductor* or semiconductor or electro* or optic* or hydrogen or superconductor* or catalyst* or thermoelectric or ions)).ti - limit to human
Scielo (results: 153)
(ti:((ti:((ti:(anabolic)) OR (ti:(doping)) OR (ti:(performance enhancing)))) AND NOT (ti:(review)))) AND NOT (ti:((ti:(animal)) AND NOT (ti:(bovine)) AND NOT (ti:(vitro)) AND NOT (ti:(mice*)) AND NOT (ti:(pig*)) AND NOT (ti:(rat*)) AND NOT (ti:(mouse)) AND NOT (ti:(spectrometry)) AND NOT (ti:(nano*)) AND NOT (ti:(conductor*)) AND NOT (ti:(semiconductor)) AND NOT (ti:(electro*)) AND NOT (ti:(optic*)) AND NOT (ti:(hydrogen)) AND NOT (ti:(superconductor*)) AND NOT (ti:(catalyst*)) AND NOT (ti:(thermoelectric)) AND NOT (ti:(ions))))

eTable 2: Risk of bias

	Risk of bias: MMAT criteria*					
Authors, year	**1**	**2**	**3**	**4**	**5**	**Observations**
Yesalis et al., 1988	Y	N	?	Y	Y	Small sample of AAS users. Source of information described as 'Physician/pharmacist/medical literature'.
Johnson et al., 1989	Y	Y	Y	Y	Y	n/a
Kisling et al., 1989	Y	Y	?	Y	Y	Nonresponse rate not informed.
Lindstrom et al., 1990	Y	Y	Y	Y	N	The number of AAS users seeking a physician must be inferred from the text.
Terney et al., 1990	Y	Y	N	Y	Y	The number of AAS users seeking support from physicians was extrapolated from the total number of participants.
Peters et al., 1993	Y	Y	Y	?	Y	Nonresponse rate not informed.
Tanner et al., 1995	Y	Y	N	Y	Y	The number of AAS users seeking support from physicians was extrapolated from the total number of participants.
Korkia et al., 1997	Y	Y	Y	N	Y	Nonresponse rate = 41%.
Bolding et al., 1999	Y	Y	?	Y	Y	Support described as 'regular health checks to screen for steroid-related health problems'.
Auge et al., 1999	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Perry et al., 2005	Y	Y	N	Y	Y	The number of AAS users seeking a physician must be inferred from a graph and was extrapolated from the total number of participants.
Parkinson et al., 2005	Y	Y	Y	?	Y	Nonresponse rate not informed.
Pope et al., 2004	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Striegel et al., 2006	Y	Y	Y	?	Y	Nonresponse rate not informed.
Cohen et al., 2007	Y	Y	?	?	Y	Nonresponse rate not informed. Support described as 'willing to seek medical supervision'.
Al-Falasi et al., 2008	Y	N	N	N	Y	Nonresponse rate = 56%. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Larance et al., 2008	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Posiadala et al., 2010	Y	N	Y	N	?	Small sample of AAS users. Nonresponse rate = 28.57%.
Gradidge et al., 2011	Y	Y	N	Y	Y	Small sample of AAS users. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Ip et al., 2011	Y	Y	Y	Y	Y	n/a
Santos et al., 2011	Y	N	N	?	Y	Small sample of AAS users. Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Hope et al., 2013	Y	Y	Y	Y	Y	n/a
Raschka et al., 2013	Y	N	N	?	Y	Small sample of AAS users. Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Mooney et al., 2016	Y	N	N	?	Y	Small sample of AAS users. Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Rowe et al., 2016	Y	Y	Y	Y	Y	n/a
Westerman et al., 2016	Y	Y	Y	Y	Y	n/a
Zahnow et al., 2017	Y	Y	Y	?	Y	Nonresponse rate not informed.
Althobiti et al., 2018	Y	Y	N	?	Y	Nonresponse rate not informed. The number of AAS users seeking a physician was extrapolated from the total number of participants.
Hill et al., 2019	Y	Y	Y	Y	Y	n/a
Jacka et al., 2019	Y	Y	Y	?	Y	Nonresponse rate not informed.

Macedo et al., 2019	Y	N	Y	?	Y	Small sample of AAS users. Nonresponse rate not informed.
Pany et al., 2019	Y	Y	Y	?	Y	Nonresponse rate not informed.
Pereira et al., 2019	Y	Y	Y	?	Y	Nonresponse rate not informed.
Uddin et al., 2019	Y	Y	N	Y	Y	The number of AAS users seeking a physician must be inferred from the text.
Bonnecaze et al., 2020	Y	Y	Y	Y	Y	n/a
Jokipalo et al., 2021	Y	Y	Y	Y	Y	n/a

* MMAT's quality-assessment criteria:
1 - Is the sampling strategy relevant to address the research question?
2 - Is the sample representative of the target population?
3 - Are the measurements appropriate?
4 - Is the risk of nonresponse bias low?
5 - 'Is the statistical analysis appropriate to answer the research question?'
Y = yes. N = no. ? = not clear.

R codes for the meta-analysis, forest plots and meta-regression

The table below contains the dataset of the file 'metaAAS.csv'. The column 'ti' contains the number of androgenic-anabolic steroids (AAS) users in each study, and the column 'xi', the number of AAS users who sought support from physicians. The subsequent columns contain the coded variables utilised in the forest plots and meta-regression.

study id	studies	ti	xi	locat	pop	size	time	bias	design
1	Yesalis et al.,1988	15	8	0	1	0	0	0	0
2	Johnson et al.,1989	95	28	0	0	0	0	1	0
3	Kisling et al.,1989	85	21	2	3	0	0	1	0
4	Lindstrom et al.,1990	138	12	2	1	1	0	1	0
5	Terney et al.,1990	94	5	0	0	0	0	1	0
6	Peters et al.,1993	100	42	3	3	0	0	1	0
7	Tanner et al.,1995	184	33	0	0	1	0	1	0
8	Korkia et al.,1997	110	39	1	3	1	0	1	0
9	Bolding et al.,1999	81	25	1	3	0	0	1	0
10	Auge et al.,1999	17	8	0	1	0	0	0	1
11	Perry et al.,2005	207	46	0	1	1	1	1	0
12	Parkinson et al.,2005	500	185	6	3	1	1	1	2
13	Pope et al.,2004	43	16	0	1	0	1	0	1
14	Striegel et al.,2006	84	47	2	3	0	1	1	0
15	Cohen et al.,2007	1955	1290	0	3	2	1	0	2
16	Al-Falasi et al.,2008	34	4	5	3	0	1	0	0
17	Larance et al.,2008	60	46	3	3	0	1	0	1
18	Posiadala et al.,2010	18	2	2	3	0	1	0	0
19	Gradidge et al.,2011	4	1	5	0	0	1	1	0
20	Ip et al.,2011	506	387	6	3	1	1	1	2
21	Santos et al.,2011	41	4	4	1	0	1	0	0
22	Hope et al.,2013	395	178	1	2	1	1	1	0
23	Raschka et al.,2013	79	30	2	3	0	1	0	0
24	Mooney et al.,2016	26	1	1	3	0	1	0	2
25	Rowe et al.,2016	605	382	3	2	1	1	1	0
26	Westerman et al.,2016	231	153	6	3	1	1	1	2
27	Zahnow et al.,2017	195	68	6	3	1	1	1	2
28	Althobiti et al.,2018	476	181	5	3	1	1	0	0
29	Hill et al.,2019	216	91	1	1	1	1	1	2
30	Jacka et al.,2019	267	237	3	3	1	1	1	2
31	Macedo et al.,2019	25	9	4	3	0	1	0	0
32	Pany et al.,2019	74	24	5	1	0	1	1	0
33	Pereira et al.,2019	194	117	4	3	1	1	1	0
34	Uddin et al.,2019	512	9	5	3	1	1	1	0
35	Bonnecaze et al.,2020	2385	1047	6	3	2	1	1	2
36	Jokipalo et al.,2021	50	15	2	1	0	1	1	2

R codes

```
## Loading packages previously installed
library("robumeta")
library("metafor")
library("dplyr")
library("forestplot")
library("checkmate")
library("nloptr")
library("meta")
library("CompQuadForm")

## Meta-analysis of all studies
metaAAS <- read.csv("metaAAS.csv")
View(metaAAS)
metaAAS <- escalc(measure="IR", xi=xi, ti=ti, data=metaAAS, slab=paste (studies))
res <- rma(yi, vi, data=metaAAS)
res
confint(res)

## Baujat plot
b_res <- rma(xi, ti, data=metaAAS, slab=study.id)
baujat(b_res)
inf <- influence(res)
print(inf)
plot(inf)


### funnel plot
funnel(res,atransf=mytransf,xlab = "Estimated prevalence")
regtest(res)
ranktest(res)
warnings()


### Cross-tabulation of the number of studies – location vs subpopulation
addmargins(table(metaAAS$locat, metaAAS$pop))


### Setup functions for the forest plot
mlabfun <- function(text, res) {
 list(bquote(paste(.(text),
            " (Q = ", .(formatC(res$QE, digits=2, format="f")),
            ", df = ", .(res$k - res$p),
            ", p ", .(metafor:::.pval(res$QEp, digits=2, showeq=TRUE, sep=" ")), "; ",
            I^2, " = ", .(formatC(res$I2, digits=1, format="f")), "%, ",
            tau^2, " = ", .(formatC(res$tau2, digits=2, format="f")), ")")))}
mytransf<-function(x)
 (x)*100


## Forest plot of all studies
forest (res, transf=mytransf, cex=.7,header=c("Authors and Year", "Effect size [95%
CI]"),xlab="Prevalence rate",
     mlab=mlabfun("RE Model for All Studies", res), order="obs", refline=NA)
```

```
## Forest plot - subgroup LOCATION
forest (res, transf=mytransf, cex=.6, refline=NA,top=0,
      order=metaAAS$locat,
      rows=c(65:58,53:49,44:39,34:31,26:24,19:15,10:6),
      mlab=mlabfun("RE Model for All Studies", res),
      header=c("Authors and year", "Effect size [95% CI]"),
      xlab="Estimated prevalence")


### switch to bold italic font
par(font=4)

### add text for the subgroups
text(-130, c(54,45,35,27,20,11), pos=4,
    c("UK","Continental Europe","Australia","Brazil",
     "Africa,Asia and Middle East","Trans-regional"))

### fit random-effects model in the seven subgroups
res.0 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="0"), data=metaAAS)
res.1 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="1"), data=metaAAS)
res.2 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="2"), data=metaAAS)
res.3 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="3"), data=metaAAS)
res.4 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="4"), data=metaAAS)
res.5 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="5"), data=metaAAS)
res.6 <- rma(measure="IR", xi=xi, ti=ti, subset=(locat=="6"), data=metaAAS)

### add summary polygons for the seven subgroups
addpoly(cex=.5,res.0, row=56.5, mlab=mlabfun("RE Model for Subgroup", res.0),transf=mytransf)
addpoly(cex=.5,res.1, row=47.5, mlab=mlabfun("RE Model for Subgroup", res.1),transf=mytransf)
addpoly(cex=.5,res.2, row=37.5, mlab=mlabfun("RE Model for Subgroup", res.2),transf=mytransf)
addpoly(cex=.5,res.3, row=29.5, mlab=mlabfun("RE Model for Subgroup", res.3),transf=mytransf)
addpoly(cex=.5,res.4, row=22.5, mlab=mlabfun("RE Model for Subgroup", res.4),transf=mytransf)
addpoly(cex=.5,res.5, row=13.5, mlab=mlabfun("RE Model for Subgroup", res.5),transf=mytransf)
addpoly(cex=.5,res.6, row=4.5, mlab=mlabfun("RE Model for Subgroup", res.6),transf=mytransf)


### fit meta-regression model to test for subgroup differences
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ factor(locat), data=metaAAS)
resMeta
confint(resMeta)

### add text for the test of subgroup differences
text(-130, -3, pos=4, cex=0.6,
    bquote(paste("Test of moderator (location): ",Q[M], " = ",
            .(formatC(resMeta$QM, digits=2, format="f")),
            ", df = ", .(resMeta$p - 1),
                        ", p = ", .(formatC(resMeta$QMp, digits=2, format="f")))))
```

```
##Forest plot -subgroup SUBPOPULATION
forest (res, transf=mytransf, cex=.6, refline=NA,top=0,
      order=metaAAS$pop,
      rows=c(53:50,45:37,32:31,26:6),
      mlab=mlabfun("RE Model for All Studies", res),
      header=c("Authors and year", "Effect size [95% CI]"),
      xlab="Estimated prevalence")

### fit random-effects model in the four subgroups
res.0 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="0"), data=metaAAS)
res.1 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="1"), data=metaAAS)
res.2 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="2"), data=metaAAS)
res.3 <- rma(measure="IR", xi=xi, ti=ti, subset=(pop=="3"), data=metaAAS)

### add summary polygons for the four subgroups
addpoly(cex=.5,res.0, row=48, mlab=mlabfun("RE Model for Subgroup", res.0),transf=mytransf)
addpoly(cex=.5,res.1, row=35, mlab=mlabfun("RE Model for Subgroup", res.1),transf=mytransf)
addpoly(cex=.5,res.2, row=29, mlab=mlabfun("RE Model for Subgroup", res.2),transf=mytransf)
addpoly(cex=.5,res.3, row=4, mlab=mlabfun("RE Model for Subgroup", res.3),transf=mytransf)


### fit meta-regression model to test for subgroup differences
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ factor(pop), data=metaAAS)
resMeta
confint(resMeta)

### add text for the test of subgroup differences
text(-130, -3, pos=4, cex=0.6,
   bquote(paste("Test of moderator (sub-population): ",Q[M], " = ",
            .(formatC(resMeta$QM, digits=2, format="f")),
            ", df = ", .(resMeta$p - 1),
            ", p = ", .(formatC(resMeta$QMp, digits=2, format="f")))))


### Univariate meta-regression model. Moderator: LOCATION
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(locat), ref="0"),  data=metaAAS)
resMeta
confint(resMeta,transf=mytransf)


### Univariate meta-regression model. Moderator: SUBPOPULATION
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(pop), ref="3"), data=metaAAS)
resMeta
confint(resMeta)

### Univariate meta-regression model. Moderator: SAMPLE SIZE
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(size), ref="0"), data=metaAAS)
resMeta
confint(resMeta)

### Univariate meta-regression model. Moderator: TIME OF PUBLICATION
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(time), ref="1"), data=metaAAS)
resMeta
confint(resMeta)
```

Univariate meta-regression model. Moderator: STUDY DESIGN

```
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(design), ref="1"),
data=metaAAS)
resMeta
confint(resMeta)
```

Univariate meta-regression model. Moderator: RISK OF BIAS

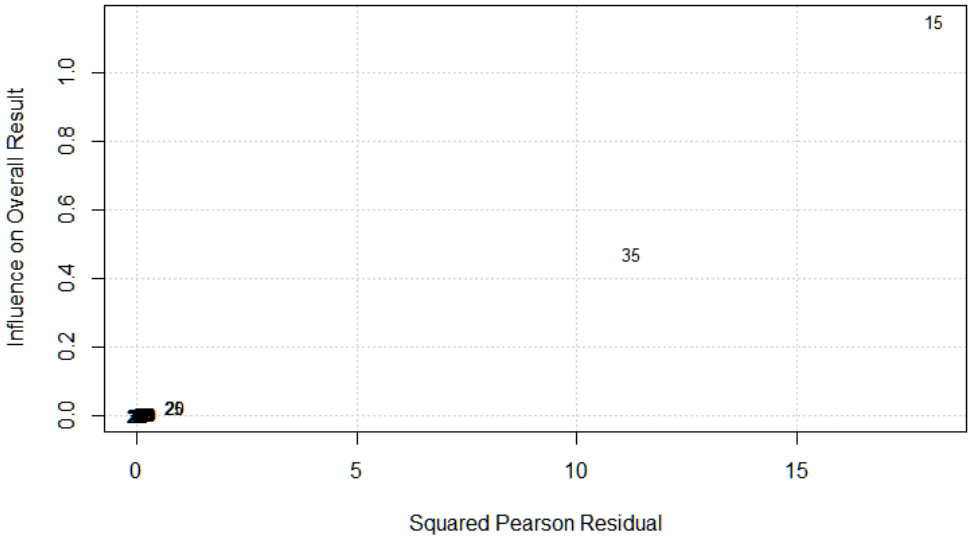
```
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ relevel(factor(bias), ref="1"), data=metaAAS)
resMeta
confint(resMeta)
```

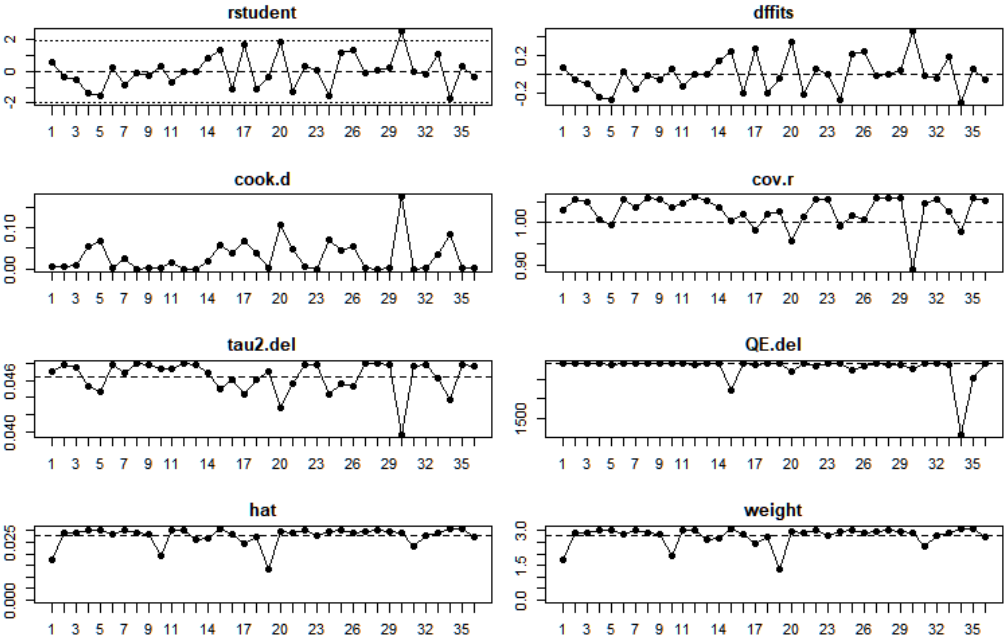
Multivariate meta-regression model. Moderators: LOCATION + STUDY DESIGN

```
resMeta <- rma(measure="IR",xi=xi, ti=ti, mods = ~ factor(locat)+relevel(factor(design),
ref="0"), data=metaAAS)
resMeta
confint(resMeta)
```

Results of the Baujat plot

	rstudent	dffits	cook.d	cov.r	tau2.del	QE.del	hat	weight	dfbs	inf
Yesalis et al.,1988	0.5691	0.0751	0.0057	1.0296	0.0470	2901.5926	0.0174	1.7416	0.0749	
Johnson et al.,1989	-0.3444	-0.0615	0.0039	1.0562	0.0477	2901.1899	0.0288	2.8822	-0.0615	
Kisling et al.,1989	-0.5615	-0.0986	0.0099	1.0507	0.0475	2903.7126	0.0289	2.8937	-0.0986	
Lindstrom et al.,1990	-1.3464	-0.2365	0.0546	1.0080	0.0453	2888.7020	0.0303	3.0336	-0.2363	
Terney et al.,1990	-1.5188	-0.2662	0.0681	0.9939	0.0446	2872.4020	0.0304	3.0377	-0.2658	
Peters et al.,1993	0.2168	0.0346	0.0012	1.0567	0.0478	2891.9587	0.0282	2.8198	0.0346	
Tanner et al.,1995	-0.8914	-0.1576	0.0250	1.0380	0.0468	2904.9166	0.0301	3.0115	-0.1576	
Korkia et al.,1997	-0.0751	-0.0154	0.0002	1.0593	0.0479	2896.1842	0.0288	2.8752	-0.0154	
Bolding et al.,1999	-0.2797	-0.0501	0.0026	1.0566	0.0478	2901.0530	0.0284	2.8416	-0.0501	
Auge et al.,1999	0.3664	0.0502	0.0025	1.0367	0.0473	2902.0604	0.0193	1.9265	0.0501	
Perry et al.,2005	-0.6882	-0.1224	0.0152	1.0477	0.0473	2903.7785	0.0301	3.0053	-0.1225	
Parkinson et al.,2005	-0.0059	-0.0036	0.0000	1.0626	0.0480	2858.4148	0.0303	3.0265	-0.0036	
Pope et al.,2004	0.0035	-0.0017	0.0000	1.0536	0.0478	2900.9937	0.0259	2.5919	-0.0017	
Striegel et al.,2006	0.8253	0.1366	0.0188	1.0357	0.0468	2884.0498	0.0269	2.6891	0.1365	
Cohen et al.,2007	1.3765	0.2473	0.0594	1.0032	0.0451	2203.3151	0.0305	3.0526	0.2470	
Al-Falasi et al.,2008	-1.1563	-0.1977	0.0388	1.0208	0.0460	2903.5873	0.0286	2.8616	-0.1977	
Larance et al.,2008	1.6754	0.2655	0.0680	0.9820	0.0444	2878.6129	0.0241	2.4114	0.2664	
Posiadala et al.,2010	-1.1536	-0.1919	0.0366	1.0200	0.0461	2904.0446	0.0271	2.7140	-0.1920	
Gradidge et al.,2011	-0.3687	-0.0438	0.0019	1.0253	0.0470	2904.9093	0.0131	1.3108	-0.0436	
Ip et al.,2011	1.9001	0.3414	0.1077	0.9560	0.0428	2681.1281	0.0298	2.9778	0.3407	
Santos et al.,2011	-1.2671	-0.2186	0.0470	1.0136	0.0456	2901.6068	0.0292	2.9249	-0.2186	
Hope et al.,2013	0.3645	0.0618	0.0039	1.0575	0.0477	2842.9150	0.0300	3.0010	0.0619	
Raschka et al.,2013	0.0375	0.0040	0.0000	1.0575	0.0479	2897.1920	0.0279	2.7864	0.0040	
Mooney et al.,2016	-1.5763	-0.2731	0.0714	0.9891	0.0444	2889.9271	0.0298	2.9798	-0.2728	
Rowe et al.,2016	1.2229	0.2170	0.0463	1.0150	0.0457	2712.0790	0.0301	3.0072	0.2168	
Westerman et al.,2016	1.3474	0.2350	0.0539	1.0053	0.0452	2825.5656	0.0290	2.8960	0.2349	
Zahnow et al.,2017	-0.1027	-0.0204	0.0004	1.0610	0.0479	2890.1445	0.0296	2.9607	-0.0204	
Althobiti et al.,2018	0.0413	0.0047	0.0000	1.0624	0.0480	2856.9754	0.0302	3.0228	0.0047	
Hill et al.,2019	0.2275	0.0373	0.0014	1.0592	0.0479	2876.4907	0.0295	2.9508	0.0373	
Jacka et al.,2019	2.5277	0.4496	0.1749	0.8926	0.0397	2756.3671	0.0287	2.8694	0.4492	
Macedo et al.,2019	-0.0457	-0.0092	0.0001	1.0485	0.0476	2902.8855	0.0235	2.3471	-0.0091	
Pany et al.,2019	-0.2084	-0.0378	0.0015	1.0570	0.0478	2900.6346	0.0281	2.8096	-0.0378	
Pereira et al.,2019	1.0596	0.1829	0.0333	1.0252	0.0462	2848.8713	0.0288	2.8819	0.1829	
Uddin et al.,2019	-1.7165	-0.3019	0.0858	0.9758	0.0437	1099.7502	0.0307	3.0725	-0.3012	
Bonnecaze et al.,2020	0.3139	0.0533	0.0029	1.0598	0.0478	2520.6043	0.0306	3.0626	0.0534	
Jokipalo et al.,2021	-0.3118	-0.0543	0.0030	1.0537	0.0477	2902.8279	0.0272	2.7230	-0.0543	


Influence on Overall Result
Squared Pearson Residual
15
35
20
0.0
0.2
0.4
0.6
0.8
1.0
0
5
10
15


rstudent
dffits
cook.d
cov.r
tau2.del
QE.del
hat
weight